

CST3144 WEEK 2 LAB TASKS

TASK 1. (COURSEWORK TASK) VERSION CONTROL WITH GITHUB:

- i. Check if Git is already installed in your computer (in the terminal, type `git -v`). If it is not, install it from the Git website: <https://git-scm.com/downloads> .
- ii. Check if Git is configured with your name and email r (in the terminal, type `git config --list`). If not, setup your name and email with the following commands (replacing the name and email with your own) :
`git config --global user.name "Your Name"`
`git config --global user.email your@email.address`
- iii. Create an account on <https://github.com/>
- iv. Create a public repository to host the front-end of your coursework project.
On the page for this repository, click the 'code button' and copy the 'clone using URL' link.
Now in Visual Studio Code, open a new window and click 'Clone Git Repository'. When prompted, provide the clone link URL of your project. Choose a folder on your computer to sort the local repository.
Copy the source files for your coursework project into the local repository.
Alternatively, you may open your existing coursework project on your computer and use it to create a new public repository on github (click on the Version Control tab and click 'Publish to Github'). When prompted, provide the login details for your github account to link it to VSC.
- v. Make at least one edit to your project (e.g. changing the price) and check that your webpage still displays correctly on your computer (using Live Preview or Live Server).
- vi. In the Version Control tab, stage your changes: click the '+' icon next to new or modified files. Above the 'sync changes' button, add a short commit message that describes your changes. Then click 'sync changes' to push your changes to the github repository. When prompted, provide the login details for your github account to link it to VSC
- vii. On github, check that your most recent changes appear I the commit history

TASK 2 (COURSEWORK TASK) MAKING THE FRONT-END PUBLICALLY AVAILABLE:

- viii. On Github, go to the setting tab for your project, then 'Pages'. For "Branch", select "main" and click "Save".
- ix. After a few moments (and reloading the page), you should have a link to the GitHub pages address for your project.
- x. Go to this address and check that your webpage looks correct.

Notes:

The html page should be called index.html or it may not load properly.

If some files (e.g. images) are not loading correctly, check that the source address is relative to the project folder (e.g. `"/images/computerScience.jpg"` not

`"C:/users/myname/Documents/CST3144/courseworkproject/imates/scomputerScienc e.jpg"`)

- xi. **For future coursework tasks, make sure to push your changes to github.**

TASK 3 INTERACTIVITY PRACTICE

This task is not directly related to the coursework, but the concepts covered here will help you with the tasks that follow.

From the module page, download the file 'interactivityExample.html' and open it in VSC. Open the webpage in Live Preview / Live Server and see how it works.

- Observe that the 'Get More Email' button is disabled because of the following directive : `v-bind:disabled="true"`. Change this to `false` so that the button is enabled.
- Observe that the 'Read An Email' button calls the method `readEmail()` when clicked. This method in turn decreases the value of `unreadEmails` by 1.
- Add a method `getNewEmail` that increases value of `unreadEmails` by 5.
- Bind the `getNewEmail` method to the 'Get More Email' button, so that when this button is clicked the number of unread emails goes up.
- Observe that `fullname` is a computed value that depends on the values of `firstName` and `lastName`. Create a new computed value called `totalEmails` that is equal to the sum of read and unread emails. Update the welcome message to correctly display the total number of emails.
- Observe that we can keep clicking "Read An Email" even when the number of unread emails is equal to (or less than!) 0. Create a new computed value called `canReadEmail` that returns `true` if the number of unread emails is greater than 0, and returns `false` otherwise. Use this computed value to disable the "Read An Email" button when there are no unread Emails. (hint: use `v-bind:disabled`, and recall that `!` in front of a Boolean variable negates it.)

TASK 4 (COURSEWORK TASK) 'ADD TO CART' FUNCTIONALITY:

In your coursework project:

- Add an 'availability' field to your lesson that tracks how many slots are available for purchase. Give the lesson an ID number (e.g. 1001).
- Add a 'cart' array that will store the ID's of lessons added to the basket
- Create an 'add to cart' button.
- Create a method called 'addToCart' that when called:
 - Pushes a copy of the lesson id to the cart;
 - Reduces the 'availability' of the lesson by 1.Bind this method to the 'add to cart' button.
- Create a computed value called 'canAddToCart' that returns `true` if the lesson availability is greater than 0, and returns `false` otherwise.
- Edit the 'addToCart' button so that it is only enabled when 'canAddToCart' is true.

Make sure at each step to check that your webpage is still working as intended. When the task is completed, remember to push your changes to github!

TASK 5 (COURSEWORK TASK) 'SHOPPING CART' FUNCTIONALITY:

- Create a computed value called 'numberOfItems' that returns the length of the cart array.
- Add a 'Checkout' button that displays the number of items in the cart

- Create a new Boolean data variable called 'displayCart', set to false.
- By adding an appropriate method, make it so that the 'Checkout' toggles the value of displayCart.
- Using v-if and v-else, edit your html so that the lesson information is only displayed when 'displayCart' is false, and otherwise a placeholder "checkout" appears.
- Edit the 'displayCart' button so that it is only enabled when there is at least one item in the cart.

Make sure at each step to check that your webpage is still working as intended.

When the task is completed, remember to push your changes to github!

Coursework Checklist:

After completing these tasks, your coursework project should satisfy the following:

(Text in **bold** indicates tasks covered in this lab's exercises)

Front-End:

- [Display of Lesson] The following information is displayed (about one lesson): Subject, Location, Price, **Spaces (or availability: this indicates how many spaces are left)**, an Image (or Font Awesome icon) .
- ["Add to Cart" functionality]
 - **the lesson must have an "Add to Cart" button.**
 - **the button is always visible, and only enabled when space is larger than 0 .**
 - **clicking the button once (related interactions implemented by using v-on) will add one space to the shopping cart, reducing the remaining space by one.**
 - **once there is no more space, i.e. space = 0, the "Add to cart" button should be disabled but still visible, i.e. clicking it will not further reduce "space" nor add lessons to the cart.**
- ["Shopping Cart" functionality]:
 - **the shopping cart button should only be enabled after at least one lesson is added to cart.**
 - **clicking the shopping cart button should show the cart page, and clicking the button again goes back to the lesson page .**

Hosting:

- The Front-End is hosted on GitHub pages

BONUS TASKS:

Have a look at some examples of other types of events that can be bound to methods using v-on:

https://www.w3schools.com/vue/vue_v-on.php and try some of them out.

Using icons from FontAwesome <https://fontawesome.com/v4/icons/> , create a 'like' button for a social media post. The Vue data should store a number of likes the post already has (e.g. 100) and a Boolean for whether the user likes the post. The page should display a heart icon (or other icon) and

the number of likes the post currently has. The user can click on the icon to like / unlike the post. Doing this should (a) change the total number of likes displayed, and (b) change the appearance of or colour of the icon. (You can use the following to link to the icons sheet for FontAwesome v4: `<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/4.7.0/css/font-awesome.min.css">`)